

# ST. XAVIER'S COLLEGE

MAITIGHAR, KATHMANDU, NEPAL

Phone: 01-5321365, 01-5344636

Email: ktm@sxc.edu.np



LAB ASSIGNMENT NUMBER: 4.2

## Spreadsheet Software

| Submitted By                                                         | Submitted To                                                              | Signature |
|----------------------------------------------------------------------|---------------------------------------------------------------------------|-----------|
| Name: Kushma Gautam<br>Roll No: 026NEB952<br>Class: XI<br>Section: I | Department of Computer Science,<br>NEB (Plus Two)<br>St. Xavier's College |           |

Submission Date: 28<sup>th</sup> of August, 2026

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# Cell in Microsoft Excel

Define a **cell** in Microsoft Excel.

An Excel cell is the fundamental grid element formed by the intersection of a unique column and row, serving as a distinct addressable unit that stores, calculates, and formats data, text, or functional expressions.

## Labelled Blueprint of Microsoft Excel

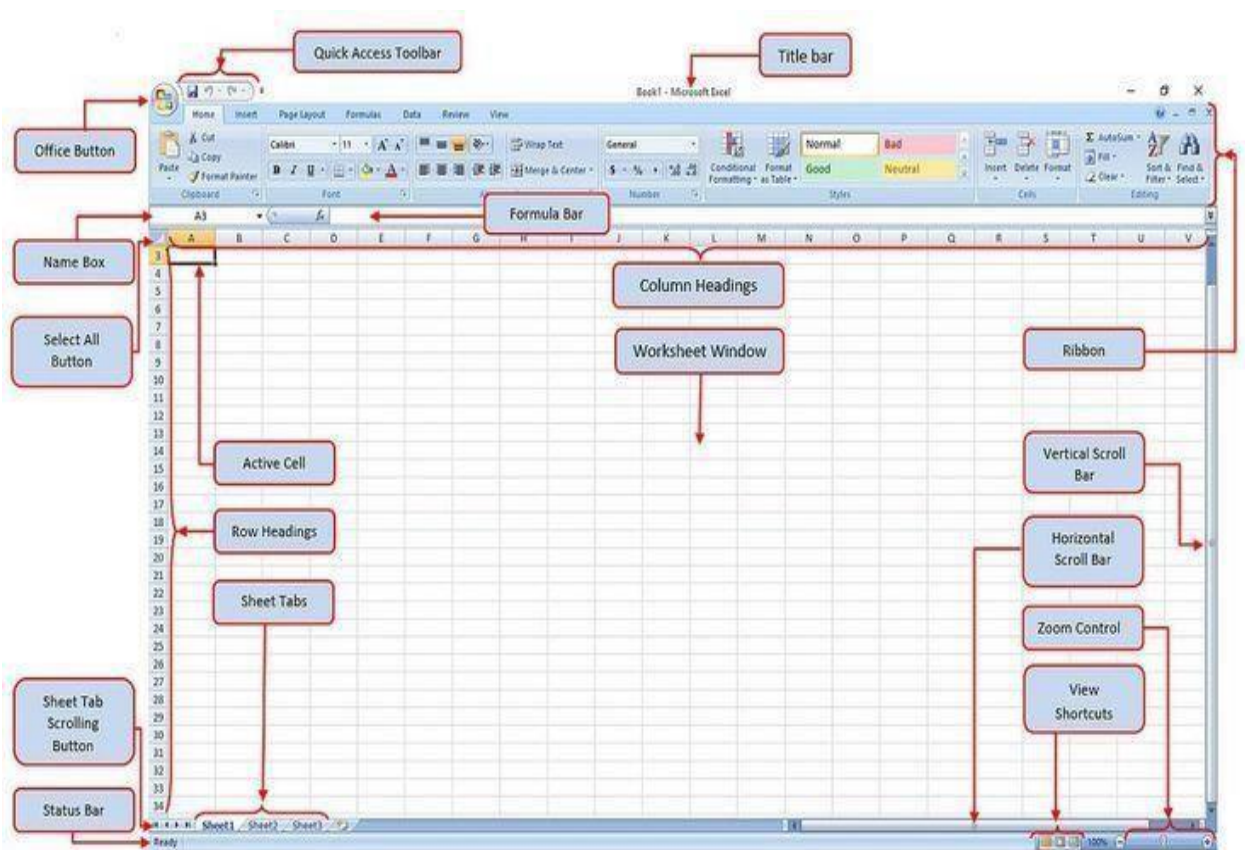


Figure 1: Structural Blueprint of the Microsoft Excel User Interface

## Cell Reference in Excel

A cell reference in Microsoft Excel is an alphanumeric address that identifies the exact location of a cell or range of cells within a worksheet, allowing formulas and functions to dynamically locate and read its stored data. There are three types of cell reference in Excel. They are:

- Relative Cell Reference
- Mixed Cell Reference
- Absolute Cell Reference

Define **Relative Cell Reference** with one example.

A relative cell reference is a standard spreadsheet reference that changes dynamically relative to the new location when copied or filled into another cell.

- **Example:** If cell **C1** contains the formula `=A1+B1` and is copied down into cell **C2**, the reference automatically updates to `=A2+B2`.

Define **Mixed Cell Reference** with one example.

A mixed cell reference is a hybrid reference that locks either only the column or only the row using a single dollar sign (\$), allowing one part of the address to remain fixed while the other part adjusts relative to the new cell location.

- **Example:** If cell **C1** contains the formula `=A1*$B1` and is copied right into cell **D1**, the formula changes to `=B1*$B1`.

Define **Absolute Cell Reference** with one example.

An absolute cell reference is a fixed spreadsheet reference that remains locked to a specific cell address using dollar signs (\$) before both the column letter and row number, preventing it from changing when copied or filled to other cells.

- **Example:** If cell **C1** contains the formula `=A1*$B$1` and is copied down into cell **C2**, the reference updates to `=A2*$B$1`, keeping the reference to cell **B1** unchanged.

# VLOOKUP

Define **VLOOKUP**.

VLOOKUP is a built-in Excel function that searches vertically down the first column of a table for a specified lookup value and returns a corresponding value from a designated column in the same row.

Write the **Syntax of VLOOKUP** and explain each argument with an example.

Syntax :

=VLOOKUP (lookup\_value, table\_array, col\_index\_num, [range\_lookup])

Arguments Breakdown :

- lookup\_value (Required): The specific value you want to search for in the very first column of your table.
- table\_array (Required): The range of cells that contains both the column you are searching in and the column containing the return value.
- col\_index\_num (Required): The column number within the selected table\_array (starting at 1 for the leftmost column) from which to retrieve the matching data.
- range\_lookup (Optional): A logical value (TRUE or FALSE) specifying the match type:
  - FALSE (or 0): Exact match .
  - TRUE (or 1): Approximate match.

Example:

To find the Department of Employee ID 102:

=VLOOKUP(102, A2:C3, 3, FALSE)

- 102 (lookup\_value): The ID you want to look up.
- A2:C3 (table\_array): The dataset range to search within (Column A is the first column).
- 3 (col\_index\_num): Returns data from the 3rd column of the range (Department).
- FALSE (range\_lookup): Demands an exact match for ID 103.

Result: Marketing

|   | A      | B     | C          |
|---|--------|-------|------------|
| 1 | Emp ID | Name  | Department |
| 2 | 101    | Sarah | Sales      |
| 3 | 102    | David | Marketing  |

| ST. XAVIER'S COLLEGE |                   |         |            |            |              |              |          |          |             |             |           |           |            |            |       |        |                            |    |         |
|----------------------|-------------------|---------|------------|------------|--------------|--------------|----------|----------|-------------|-------------|-----------|-----------|------------|------------|-------|--------|----------------------------|----|---------|
| Roll no              | Name              | Section | PHYSICS TH | PHYSICS PR | CHEMISTRY TH | CHEMISTRY PR | MATHS TH | MATHS PR | COMPUTER TH | COMPUTER PR | NEPALI TH | NEPALI PR | ENGLISH TH | ENGLISH PR | TOTAL | REMARK | KUSHMA GAUTAM Roll No. 952 |    |         |
| 1                    | Hari Shrestha     | G       | 60         | 23         | 43           | 23           | 56       | 23       | 40          | 47          | 60        | 23        | 60         | 23         | 481   | PASS   | 80.17                      | A  | 2       |
| 2                    | Arshi Dhungana    | H       | 60         | 21         | 65           | 21           | 73       | 23       | 44          | 43          | 60        | 23        | 45         | 24         | 502   | PASS   | 83.67                      | A  | 1       |
| 3                    | Ellie Karmacharya | I       | 71         | 23         | 71           | 23           | 45       | 24       | 43          | 41          | 56        | 23        | 34         | 21         | 475   | PASS   | 79.17                      | B+ | 3       |
| 4                    | Sanskrit BK       | G       | 45         | 24         | 29           | 24           | 22       | 23       | 45          | 34          | 34        | 23        | 56         | 19         | 378   | FAIL   | 63                         | NG | NO RANK |
| 5                    | Swashree Acharya  | H       | 23         | 19         | 54           | 19           | 57       | 21       | 46          | 45          | 29        | 23        | 65         | 23         | 424   | FAIL   | 70.67                      | NG | NO RANK |
| 6                    | Anuska Karna      | I       | 62         | 20         | 36           | 20           | 59       | 23       | 23          | 34          | 45        | 23        | 62         | 19         | 426   | PASS   | 71                         | B+ | 6       |
| 7                    | Liza Mainali      | I       | 68         | 24         | 27           | 24           | 34       | 22       | 46          | 26          | 68        | 21        | 56         | 23         | 439   | PASS   | 73.17                      | B+ | 5       |
| 8                    | Aayam Shilpakar   | H       | 22         | 24         | 33           | 24           | 50       | 19       | 39          | 34          | 45        | 24        | 67         | 19         | 400   | FAIL   | 66.67                      | NG | NO RANK |
| 9                    | Aryan Basnet      | G       | 69         | 24         | 46           | 24           | 29       | 23       | 34          | 22          | 34        | 21        | 43         | 17         | 386   | PASS   | 64.33                      | B  | 9       |
| 10                   | Birendra Waiba    | H       | 33         | 20         | 67           | 20           | 23       | 21       | 29          | 29          | 68        | 23        | 52         | 23         | 446   | FAIL   | 74.33                      | NG | NO RANK |

Figure 2: Screenshot of Sheet 1: Student Marksheet Calculation

[Screenshots of formula for calculation are added at the end as per the list of figures.]



## ST.XAVIER'S COLLEGE

Maitighar, Kathmandu, Nepal  
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### Report Card

Examination: First Terminal Examination

Issued Date: 28th of August, 2026

Roll No:

Name :

Section :

| S.N. | Subjects       | Obtained marks | Full marks | Sub Mark | Remarks | Grade | GPA |
|------|----------------|----------------|------------|----------|---------|-------|-----|
| 1    | Physics TH     |                | 75         |          |         |       |     |
| 2    | Physics PR     |                | 25         |          |         |       |     |
| 3    | Chemistry TH   |                | 75         |          |         |       |     |
| 4    | Chemistry PR   |                | 25         |          |         |       |     |
| 5    | English TH     |                | 75         |          |         |       |     |
| 6    | English PR     |                | 25         |          |         |       |     |
| 7    | Nepali TH      |                | 75         |          |         |       |     |
| 8    | Nepali PR      |                | 25         |          |         |       |     |
| 9    | Mathematics TH |                | 75         |          |         |       |     |
| 10   | Mathematics PR |                | 25         |          |         |       |     |
| 11   | Computer TH    |                | 50         |          |         |       |     |
| 12   | Computer PR    |                | 50         |          |         |       |     |
|      |                | Total marks    |            |          |         |       |     |
|      |                | Percentage     |            |          |         |       |     |
|      |                | Rank           |            |          |         |       |     |
|      |                | Grade          |            |          |         |       |     |
|      |                | GPA            |            |          |         |       |     |
|      |                | Remarks        |            |          |         |       |     |

  
Class Teacher

  
Principal

Figure 3: Screenshots of Sheet 2: VLOOKUP-based Student Result





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Issued Date: 28th of August,2026

Roll No: 1  
Name : Hari Shrestha  
Section : G

| S.N.        | Subjects       | Obtained marks | Full marks | Sub Mark | Remarks  | Grade | GPA |
|-------------|----------------|----------------|------------|----------|----------|-------|-----|
| 1           | Physics TH     | 60             | 75         | 83       | Pass     | A     | 3.6 |
| 2           | Physics PR     | 23             | 25         |          |          |       |     |
| 3           | Chemistry TH   | 43             | 75         | 66       | Pass     | B     | 2.8 |
| 4           | Chemistry PR   | 23             | 25         |          |          |       |     |
| 5           | English TH     | 60             | 75         | 83       | Pass     | A     | 3.6 |
| 6           | English PR     | 23             | 25         |          |          |       |     |
| 7           | Nepali TH      | 60             | 75         | 83       | Pass     | A     | 3.6 |
| 8           | Nepali PR      | 23             | 25         |          |          |       |     |
| 9           | Mathematics TH | 56             | 75         | 79       | Pass     | B+    | 3.2 |
| 10          | Mathematics PR | 23             | 25         |          |          |       |     |
| 11          | Computer TH    | 40             | 50         | 87       | Pass     | A     | 3.6 |
| 12          | Computer PR    | 47             | 50         |          |          |       |     |
| Total marks |                |                |            |          | 481      |       |     |
| Percentage  |                |                |            |          | 80.16667 |       |     |
| Rank        |                |                |            |          | 2        |       |     |
| Grade       |                |                |            |          | A        |       |     |
| GPA         |                |                |            |          | 3.4      |       |     |
| Remarks     |                |                |            |          | PASS     |       |     |



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Figure 4: Marksheet of Roll no. 1



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### Report Card

Examination: First Terminal Examination

Issued Date: 28th of August,2026

|           |                |
|-----------|----------------|
| Roll No:  | 2              |
| Name :    | Arshi Dhungana |
| Section : | H              |

| S.N.        | Subjects       | Obtained marks | Full marks | Sub Mark | Remarks  | Grade | GPA |
|-------------|----------------|----------------|------------|----------|----------|-------|-----|
| 1           | Physics TH     | 60             | 75         | 81       | Pass     | A     | 3.6 |
| 2           | Physics PR     | 21             | 25         |          |          |       |     |
| 3           | Chemistry TH   | 65             | 75         | 86       | Pass     | A     | 3.6 |
| 4           | Chemistry PR   | 21             | 25         |          |          |       |     |
| 5           | English TH     | 45             | 75         | 69       | Pass     | B     | 2.8 |
| 6           | English PR     | 24             | 25         |          |          |       |     |
| 7           | Nepali TH      | 60             | 75         | 83       | Pass     | A     | 3.6 |
| 8           | Nepali PR      | 23             | 25         |          |          |       |     |
| 9           | Mathematics TH | 73             | 75         | 96       | Pass     | A+    | 4   |
| 10          | Mathematics PR | 23             | 25         |          |          |       |     |
| 11          | Computer TH    | 44             | 50         | 87       | Pass     | A     | 3.6 |
| 12          | Computer PR    | 43             | 50         |          |          |       |     |
| Total marks |                |                |            |          | 502      |       |     |
| Percentage  |                |                |            |          | 83.66667 |       |     |
| Rank        |                |                |            |          | 1        |       |     |
| Grade       |                |                |            |          | A        |       |     |
| GPA         |                |                |            |          | 3.533333 |       |     |
| Remarks     |                |                |            |          | PASS     |       |     |

  
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Figure 5: Marksheet of Roll no. 2



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## Report Card

Examination: First Terminal Examination

Issued Date: 28th of August,2026

Roll No: 3  
Name : Ellie Karmacharya  
Section : I

| S.N.        | Subjects       | Obtained marks | Full marks | Sub Mark | Remarks  | Grade | GPA |
|-------------|----------------|----------------|------------|----------|----------|-------|-----|
| 1           | Physics TH     | 71             | 75         | 94       | Pass     | A+    | 4   |
| 2           | Physics PR     | 23             | 25         |          |          |       |     |
| 3           | Chemistry TH   | 71             | 75         | 94       | Pass     | A+    | 4   |
| 4           | Chemistry PR   | 23             | 25         |          |          |       |     |
| 5           | English TH     | 34             | 75         | 55       | Pass     | C+    | 2.4 |
| 6           | English PR     | 21             | 25         |          |          |       |     |
| 7           | Nepali TH      | 56             | 75         | 79       | Pass     | B+    | 3.2 |
| 8           | Nepali PR      | 23             | 25         |          |          |       |     |
| 9           | Mathematics TH | 45             | 75         | 69       | Pass     | B     | 2.8 |
| 10          | Mathematics PR | 24             | 25         |          |          |       |     |
| 11          | Computer TH    | 43             | 50         | 84       | Pass     | A     | 3.6 |
| 12          | Computer PR    | 41             | 50         |          |          |       |     |
| Total marks |                |                |            |          | 475      |       |     |
| Percentage  |                |                |            |          | 79.16667 |       |     |
| Rank        |                |                |            |          | 3        |       |     |
| Grade       |                |                |            |          | B+       |       |     |
| GPA         |                |                |            |          | 3.333333 |       |     |
| Remarks     |                |                |            |          | PASS     |       |     |

  
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Figure 6: Marksheets of Roll no. 3



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## Report Card

Examination: First Terminal Examination

Issued Date: 28th of August,2026

Roll No: 4  
Name : Sanskrit BK  
Section : G

| S.N.        | Subjects       | Obtained marks | Full marks | Sub Mark | Remarks  | Grade | GPA |
|-------------|----------------|----------------|------------|----------|----------|-------|-----|
| 1           | Physics TH     | 45             | 75         | 69       | Pass     | B     | 2.8 |
| 2           | Physics PR     | 24             | 25         |          |          |       |     |
| 3           | Chemistry TH   | 29             | 75         | 53       | Pass     | C+    | 2.4 |
| 4           | Chemistry PR   | 24             | 25         |          |          |       |     |
| 5           | English TH     | 56             | 75         | 75       | Pass     | B+    | 3.2 |
| 6           | English PR     | 19             | 25         |          |          |       |     |
| 7           | Nepali TH      | 34             | 75         | 57       | Pass     | C+    | 2.4 |
| 8           | Nepali PR      | 23             | 25         |          |          |       |     |
| 9           | Mathematics TH | 22             | 75         | 45       | Fail     | NG    | 2   |
| 10          | Mathematics PR | 23             | 25         |          |          |       |     |
| 11          | Computer TH    | 45             | 50         | 79       | Pass     | B+    | 3.2 |
| 12          | Computer PR    | 34             | 50         |          |          |       |     |
| Total marks |                |                |            |          | 378      |       |     |
| Percentage  |                |                |            |          | 63       |       |     |
| Rank        |                |                |            |          | NO RANK  |       |     |
| Grade       |                |                |            |          | NG       |       |     |
| GPA         |                |                |            |          | 2.666667 |       |     |
| Remarks     |                |                |            |          | FAIL     |       |     |

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Figure 7: Marksheet of Roll no. 4



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Examination: First Terminal Examination

Issued Date: 28th of August, 2026

Roll No: 5  
Name : Swashree Acharya  
Section : H

| S.N.        | Subjects       | Obtained marks | Full marks | Sub Mark | Remarks  | Grade | GPA |
|-------------|----------------|----------------|------------|----------|----------|-------|-----|
| 1           | Physics TH     | 23             | 75         | 42       | Fail     | NG    | 2   |
| 2           | Physics PR     | 19             | 25         |          |          |       |     |
| 3           | Chemistry TH   | 54             | 75         | 73       | Pass     | B+    | 3.2 |
| 4           | Chemistry PR   | 19             | 25         |          |          |       |     |
| 5           | English TH     | 65             | 75         | 88       | Pass     | A     | 3.6 |
| 6           | English PR     | 23             | 25         |          |          |       |     |
| 7           | Nepali TH      | 29             | 75         | 52       | Pass     | C+    | 2.4 |
| 8           | Nepali PR      | 23             | 25         |          |          |       |     |
| 9           | Mathematics TH | 57             | 75         | 78       | Pass     | B+    | 3.2 |
| 10          | Mathematics PR | 21             | 25         |          |          |       |     |
| 11          | Computer TH    | 46             | 50         | 91       | Pass     | A+    | 4   |
| 12          | Computer PR    | 45             | 50         |          |          |       |     |
| Total marks |                |                |            |          | 424      |       |     |
| Percentage  |                |                |            |          | 70.66667 |       |     |
| Rank        |                |                |            |          | NO RANK  |       |     |
| Grade       |                |                |            |          | NG       |       |     |
| GPA         |                |                |            |          | 3.066667 |       |     |
| Remarks     |                |                |            |          | FAIL     |       |     |

  
Class Teacher

  
Principal

Figure 8: Marksheet of Roll no. 5

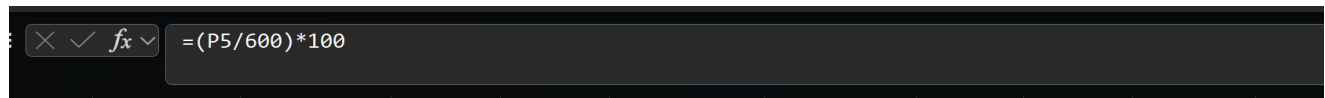


Figure 9: Screenshot of Percentage



Figure 10: Screenshot of Remark

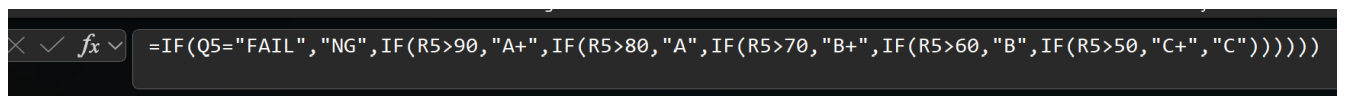


Figure 11: Screenshot of Grade



Figure 12: Screenshot of Rank



Figure 13: Screenshot of Total

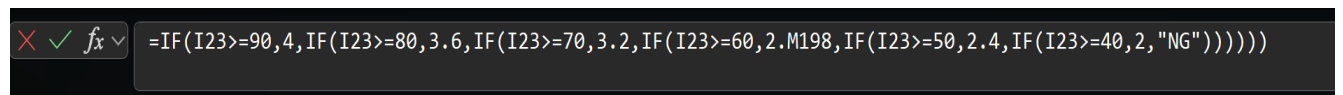


Figure 14: Screenshot of GPA

(GPA : Sheet 2 only)

## **Lab Activity Reflections**

### **Lab 1 (Fundamentals & Design):**

The first lab class was focused on the fundamentals such as learning about the Microsoft Excel User Interface and working with data entry and design (charts, illustrations, and more). With bill design and illustration-generation (pie chart, bar graph) being done as the core activity, it provided better understanding of the different features in Excel. While the procedure and foundational concepts were straightforward to follow, adjustments with cell merge and visually attractive design (bill) did require some practice.

### **Lab 2 (Formulas, Cell Referencing and VLOOKUP):**

The second lab class was primarily focused on data entry and manipulation with concept of cell references (absolute, relative and mixed), formula (text, conditional, logical, and more), and function (VLOOKUP) being introduced. While working with basic text and arithmetic formula was a familiar task, the concept of cell referencing and use of VLOOKUP remained new, that required a few attempts to understand and get done right.

### **Lab 3 (Marksheet and Report Card Design):**

The third lab class facilitated the application of formula and function so learnt with primary focus on logical formula and use of VLOOKUP through report card design. Template design was a basic task requiring cell merge and elements for visually attractive layout while data manipulation and calculation required attention to formula accuracy and remained a relatively complex task.

## **Conclusion**

This lab assignment put into effect the concepts learnt during the two lab sessions on spreadsheet software and Microsoft Excel effectively. It provided a comprehensive understanding of Excel fundamentals, including user interface, formula integration, as well as functions (VLOOKUP) and automation through tasks like worksheet data entry, calculation and report card design. Working through the assignment ingrained the idea of how efficient cell referencing and formula are for repetitive actions in Excel. With enhanced practical command over core Excel functions, visual formatting and accurate data manipulation, the completion of this project has ensured the practical application of core tools within the software.